ENDOCRINOLOGY

Lipid Abnormalities

OBJECTIVES

- Define dyslipidemia & the types of lipid abnormalities.
- Outline the secondary causes .
- Know the guidelines for the management of hyperlipidemia & its screening.

By the end of this lecture the student will be able to:

1. Define dyslipidemia & the types of lipid abnormalities.
2. Outline the secondary causes .
3. Know the guidelines for the management of hyperlipidemia & its screening.

The 3 main biological classes of lipid are:

1. cholesterol
2. Triglycerides
3. phospholipids.

Lipids are transported & metabolised by apolipoproteins.

In the fasting state , the liver is the major source of plasma lipids.

Lipids & CVD

Hyperlipidemia are major modifiable risk factor for cardiovascular diseases, increase levels of atherogenic lipoproteins (especially LDL) contribute to the development of atherosclerosis. Conversely HDL removes cholesterol from the tissues to the liver, where it is metabolised & excreted in bile.

Treating hyperlipidemia can reduce the cardiovascular events & cardiovascular mortality .

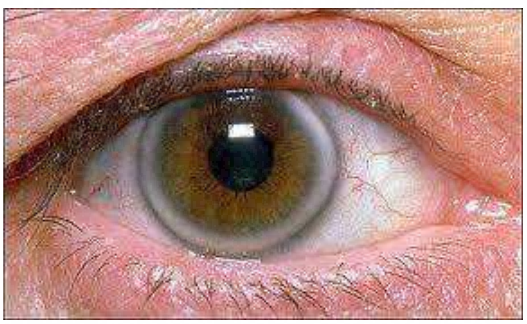
Classification of hyperlipdemia:

1. Predominant hypercholesterolemia
2. Predominant hypertriglyceridemia .
3. Mixed hyperlipidemia.

Lipid measurement are usually performed for the following reasons:

1. Screening for primary or secondary prevention of CVD.
2. Investigation of patient with clinical features of lipid disorders.
3. Monitoring of response to diet, weight control & medications.

- It is a polygenic disorder that is the most common cause of mild to moderate increase in LDL-C .
- Physical signs such as corneal arcus & xanthelasma may be found .
- The risk of CVD is proportional to the degree of LDL.
- Familial hypercholesterolemia is a more severe disorder , usually inherited as autosomal dominant , affected patients presented suffer from severe hypercholesterolemia & premature CVD.
- FH may be accompanied by xanthoma of the Achilles tendon & extensor digitorum tendon. (strongly suggest FH) , the onset of corneal arcus before age of 40 is also suggestive of this condition.




Causes of secondary hypercholesterolemia :

1. Hypothyroidism.
2. Drugs : diuretics, ciclosporin, glucocorticoids, androgens, antiretroviral d.
3. Pregnancy
4. Cholestatic liver disease.

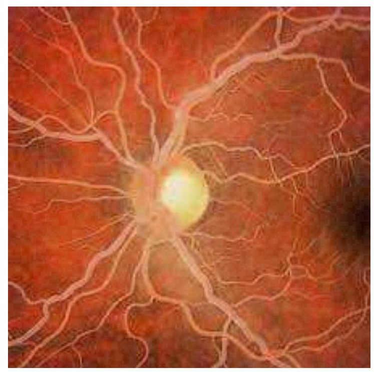
<u>**Less common:**</u>

5. Nephrotic syndrome
6. Porphyria

8. hyperparathyroidism.

Hypertriglyceridemia

- May also be polygenic , but it may be secondary to other causes (next slide).
- Familial hypertriglyceridemia can be autosomal recessive or autosomal dominant , the familial form is usually severe & can present in childhood , it is usually resistant to treatment , can lead to pancreatitis , & there is usually lipaemia retinalis , eruptive xanthoma & increased risk of CVD.



Causes of secondary hypertriglyceridemia:

1. Type 2 DM
2. Obesity
3. Chronic renal failure.
4. Excess alcohol
5. Hepatocellular disease.
6. drugs: B blockers, retinoids, glucocorticoids, antiretroviral drugs

Principles of management

- Lipid lowering therapies have a key role in the primary & secondary prevention of CVD. (what do we mean by primary & secondary prevention ?) .
- We should calculate the 10 years atherosclerotic CVD risk!.
- Public health organizations recommend target levels of lipid for patients receiving drug treatment.

a. HDL level > 38 mg /dl.
b. Fasting TG < 180 mg /dl .
c. LDL < 76 mg / dl
d. Total cholesterol <190 mg/dl.

Non pharmacological treatment

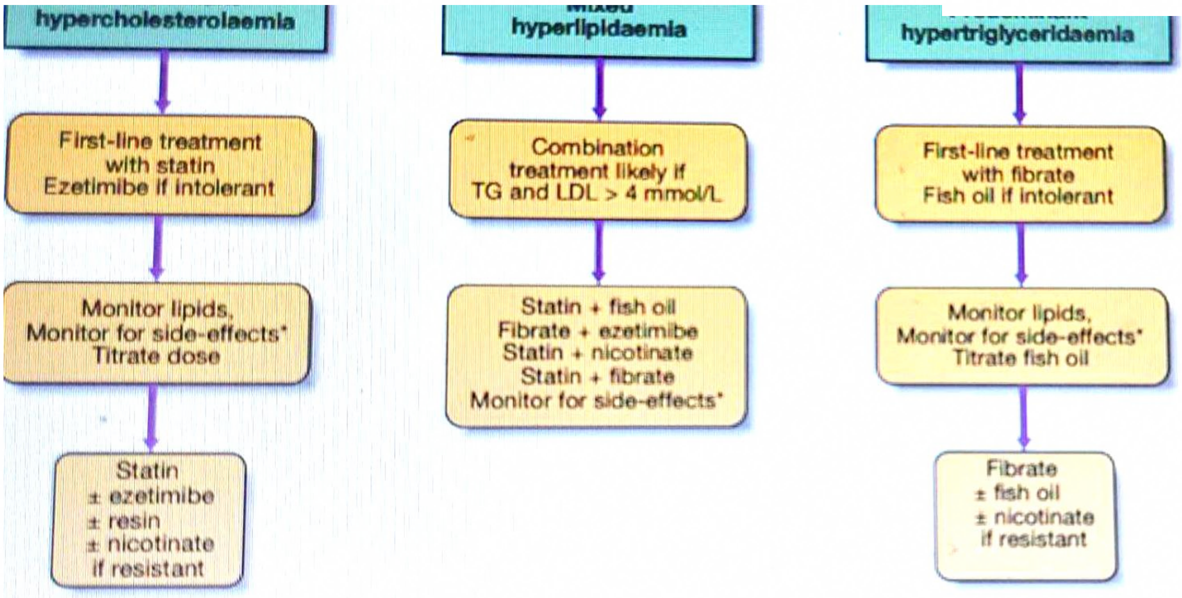
1. Reduce intake of saturated fat.
2. Reduce intake of cholesterol <250mg/day.
3. Replace sources of saturated fat & cholesterol with alternative food like lean meat , low fat dairy products & low glycemic index carb.
4. Reduce energy dense food such as fats & soft drinks .
5. Regular exercise & weight reduction.
6. Increase consumption of cardioprotective food like vegetables , fish , nuts & fruits.
7. Quit smoking & avoid alcohol.

Pharmacological treatment

Treatment of hypercholesterolemia :

Group of drugs used for hypercholesterolemia are:

1. Statins (first line)
2. Ezetimibe.
3. Bile acid sequestering agents.
4. pcsk9 inhibitors.
5. Nicotinic acid.
6. Combination of the previous agents.



Statins

- These reduce cholesterol synthesis by inhibiting the HMGCoA reductase enzyme(3-hydroxy-3-methylglutaryl-coenzyme A reductase) .
- The reduction in cholesterol synthesis up-regulates activity of the LDL receptor, which increases clearance of LDL and its precursor, IDL, resulting in a secondary reduction in LDL synthesis.
- Statins reduce LDL-C by up to 60%, reduce TG by up to 40% and increase HDL-C by up to 10%.
- There is clear evidence of protection against total and coronary mortality, stroke and cardiovascular events across the spectrum of CVD risk.
- Statins are generally well tolerated and serious side effects are rare.
- Liver function test abnormalities and muscle problems, such as myalgia, asymptomatic increase in creatine kinase (CK), myositis and, infrequently, rhabdomyolysis, are the most common.
- Side-effects are more likely in patients who are elderly, debilitated, alcoholics or receiving other drugs that interfere with statin degradation.

- Inhibits activity of the intestinal mucosal transporter NPC1L1 that absorbs dietary and biliary cholesterol.
- This mechanism of action is synergistic with the effect of statins.
- Monotherapy with the standard 10 mg/day dose reduces LDL-C by 15–20%.
- Slightly greater (17–25%) incremental LDL-C reduction occurs when ezetimibe is added to statins
- Ezetimibe is well tolerated, and evidence of a beneficial effect on cardiovascular disease endpoints is now avialable.

Bile acid sequestering resins

- Drugs in this class include colestyramine, colestipol and colesevelam.
- These prevent the reabsorption of bile acids.
- As with ezetimibe, the resultant depletion of hepatic cholesterol up-regulates LDL receptor activity and reduces LDL-C in a manner that is synergistic with the action of statins.
- Resins reduce LDL-C and modestly increase HDLC, but may increase TG.
- Colesevelam has fewer gastrointestinal effects than older preparations.

PCSK9 inhibitors

- Proportion converses subtilisin/kexin type 9 (PCSK9), an enzyme (serine protease) encoded by the PCSK9 gene, is predominantly produced in the liver.
- PCSK9 binds to the low density lipoprotein receptor (LDL-R) on the surface of hepatocytes, leading to the degradation of the LDL-R and subsequently to higher plasma LDL-cholesterol (LDL-C) levels.
- PCSK9 inhibitors are monoclonal ab interfere with the binding of the PCSK9 to LDL-R, leading to higher hepatic LDL-R expression and lower plasma LDL-C levels.
- Pcsk9 inhibitors like (alirocumab) & (evolocumab) are given by S.C injection every 2-4 weeks.
- They are well tolerated & very effective .

therapy & this has been accompanied by reduction in the risk of CVD of about 15%.

Nicotinic acid

- Pharmacological doses reduce peripheral fatty acid release with the result that VLDL and LDL decline whilst HDL-C increases.
- Randomised clinical trials have been inconsistent regarding effects on atherosclerosis and cardiovascular events.
- **Side-effects** include flushing, gastric irritation, liver function disturbances, and exacerbation of gout and hyperglycaemia.
- Slow-release formulations and low-dose aspirin may reduce flushing.

Combination therapy

- In many patients, treatment of predominant hypercholesterolaemia can be achieved by diet plus the use of a statin in sufficient doses to achieve target LDL-C levels.
- Patients who do not reach LDL targets on the highest tolerated statin dose, or who are intolerant of statins, may receive ezetimibe, nicotinic acid or resins.
- Ezetimibe and resins are safe and effective in combination with a statin, but nicotinic acid with a statin requires a greater level of caution because the risk of side-effects is slightly increased.

Treatment of hypertriglyceridemia :

Predominant hypertriglyceridemia may be treated with the followings :

1. Fibrates .
2. Highly polyunsaturated long chain n-3 fatty acid (omega3).

- These stimulate peroxisome proliferator activated receptor (PPAR) alpha, which controls the expression of gene products that mediate the metabolism of TG and HDL.
- As a result, synthesis of fatty acids, TG and VLDL is reduced.
- fibrates reduce TG by up to 50%, and increase HDL-C by up to 20%, but LDL-C changes are variable.
- Fibrates are usually well tolerated but share a similar sideeffect profile to statins. In addition, they may increase the risk of cholelithiasis and prolong the action of anticoagulants.
- Accumulating evidence suggests that they may also have a protective effect against diabetic microvascular complications.

Highly polyunsaturated long-chain n-3 fatty acids.

- These include eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), which comprise approximately 30% of the fatty acids in fish oil.
- EPA and DHA are potent inhibitors of VLDL TG formation.
- Intakes of greater than 2 g n-3 fatty acid (equivalent to 6 g of most forms of fish oil) per day lower TG in a dosedependent fashion.
- Changes in LDL-C and HDL-C are variable.

Mixed hyperlipidaemia

- Statin plus fenofibrates (pay attention to the risk of myopathy , risk is higher with gemfibrozil > fenofibrate)
- Statin plus omega 3 (if TG not too high)
- Fibrate plus ezetimibe (if cholesterol not too high) .

- The effect of drug therapy should be assessed after 6 weeks (12 weeks for fibrates).
- At this point, it is prudent to review sideeffects, lipid response, CK and liver function tests.
- It is not necessary to perform routine checks of CK and liver function unless symptoms occur, or if statins are used in combination with fibrates, nicotinic acid or other drugs that may interfere with their clearance.
- If myalgia or weakness occurs in association with CK elevation over 5–10 times the upper limit of normal, or if sustained alanine aminotransferase (ALT) elevation more than 2–3 times the upper limit of normal occurs that is not accounted for by fatty liver , treatment should be discontinued and alternative therapy sought.

References :

1. Davidson principles & practice of medicine .
2. Medscape .
3. Oxford handbook of medicine .